Rodrigo Fernandez

Senior Python AI/ML Engineer

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Summary

Senior Python AI/ML and LLM Engineer with 11+ years building cloud data products across fintech, healthcare, education, SaaS, and infrastructure industries, using **Python 3.8–3.12**, **FastAPI 0.85–0.110**, **Django 3.2–4.2**, **PyTorch 1.10–2.2**, **TensorFlow 2.x**, **scikit-learn 0.24–1.4**, **LangChain 0.1–0.2**, **OpenAI API 1.x**, **AWS SageMaker**, **Lambda**, **S3**, and **PostgreSQL 12–16** to deliver RAG search, predictive models, batch and real-time pipelines, model monitoring, API platforms, vector search, secure prompt workflows, CI/CD automation, and production LLM features with measurable reliability, cost, and user-experience improvements.

Skills

- **Languages: Python 3.8–3.12**, SQL, TypeScript, JavaScript, Bash
- **AI / ML / LLM: PyTorch 1.10–2.2**, **TensorFlow 2.x**, **scikit-learn 0.24–1.4**, XGBoost, Hugging Face Transformers, OpenAI API 1.x, LangChain 0.1–0.2, LangGraph, RAG, embeddings, prompt engineering, model evaluation, feature engineering, fine-tuning workflows
- **Backend: FastAPI 0.85–0.110**, Django 3.2–4.2, Flask 2.x, REST APIs, async processing, Celery 5.x, Pydantic, SQLAlchemy
- **Data Engineering:** pandas 1.x–2.x, NumPy, Airflow 2.x, Spark 3.x, ETL/ELT pipelines, batch jobs, event-driven pipelines, data validation
- **Vector / Search:** PostgreSQL pgvector, OpenSearch, Elasticsearch 7.x–8.x, FAISS, semantic search, hybrid search, reranking
- **Cloud / DevOps: AWS SageMaker**, Lambda, S3, ECS, EKS, API Gateway, Step Functions, SQS, SNS, CloudWatch, IAM, Docker, Kubernetes, Terraform, GitHub Actions
- **Databases:** PostgreSQL 12–16, MySQL 8.x, MongoDB 4.x–6.x, Redis 6.x–7.x, DynamoDB
- **Frontend:** React 17–18, TypeScript 4.x–5.x, dashboard integrations, AI-assisted user workflows
- **Testing / Quality:** Pytest, unit testing, integration testing, model regression testing, API contract testing, monitoring, observability

Experience

Lightedge — Senior Software Engineer *(Apr 2020 – Mar 2026)*

- Built production **LLM-powered knowledge search** using **Python 3.10–3.12**, **FastAPI 0.95–0.110**, **LangChain 0.1–0.2**, OpenAI API, PostgreSQL pgvector, and AWS services for internal analytics and customer-support workflows.
- Designed a scalable **RAG pipeline** that chunked documents, generated embeddings, stored vectors, and returned source-grounded answers, improving search relevance by **38%** and reducing repeated support investigation time.
- Implemented secure prompt templates, retrieval filters, confidence scoring, and fallback handling to reduce hallucinated answers by **31%** across compliance-sensitive business documents and operational dashboards.
- Migrated legacy Python services from **Python 3.8 to Python 3.11**, updated dependency constraints, replaced blocking I/O patterns, and improved API response latency by **27%** under peak traffic.

- Resolved a recurring production issue where malformed embeddings caused vector-search failures by adding schema validation, retry logic, dead-letter queues, and detailed CloudWatch alerts.
- Developed ML classification services with **scikit-learn 1.x**, pandas, and FastAPI to categorize tickets, detect priority signals, and route operational incidents with **24%** better assignment accuracy.
- Implemented batch and near-real-time pipelines using **AWS Lambda**, **S3**, **Step Functions**, **SQS**, and PostgreSQL to process analytics events reliably across multiple customer-facing platforms.
- Built model monitoring checks for drift, latency, token usage, failed retrievals, and API errors, giving engineering and product teams faster visibility into AI quality and production reliability.
- Created reusable Python packages for embeddings, document parsing, API clients, and model evaluation, allowing multiple teams to apply **AI/ML** features without duplicating core platform logic.
- Improved cloud cost control by optimizing token usage, caching frequent retrieval responses, pruning unused embeddings, and reducing monthly AI infrastructure spend by **22%**.
- Partnered with frontend engineers to expose AI insights through React and TypeScript dashboards, making predictions, summaries, and document evidence easier for non-technical users to review.
- Strengthened CI/CD with GitHub Actions, Docker, Pytest, Terraform checks, and automated regression tests so AI features could ship safely without breaking existing backend workflows.

NIX United —Software Engineer *(Oct 2016 – Mar 2020)*

- Developed **Python 3.6–3.8** backend and ML services for healthcare, logistics, and SaaS clients using Django, Flask, scikit-learn, TensorFlow 1.x–2.x, PostgreSQL, Redis, and AWS.
- Built predictive models for customer behavior and operational risk using feature engineering, cross-validation, and model tuning, improving forecast accuracy by **29%** for reporting workflows.
- Created REST APIs around ML models with Flask and Django REST Framework, allowing business applications to consume predictions without exposing model complexity to frontend teams.
- Migrated older Django applications from **Django 1.11 to Django 2.2**, fixed middleware compatibility issues, updated ORM queries, and reduced runtime errors after deployment.
- Solved a difficult data-quality issue where inconsistent timestamp formats broke training pipelines by adding validation rules, normalization jobs, and automated failure notifications.
- Implemented asynchronous processing with Celery, Redis, and scheduled workers to handle report generation, model scoring, and long-running data tasks without slowing customer-facing APIs.
- Improved SQL query performance by rewriting joins, adding targeted indexes, and optimizing pagination, reducing heavy analytics query time by **35%** on PostgreSQL-backed systems.
- Designed reusable data-preparation modules with pandas and NumPy so the same transformations could support dashboards, model training, and scheduled batch exports.
- Integrated AWS S3, Lambda, CloudWatch, and IAM policies into Python services to support secure file ingestion, audit-friendly processing, and operational monitoring.
- Collaborated with QA and DevOps teams to add Pytest coverage, Dockerized local environments, and CI checks that caught dependency and migration errors before release.
- Applied **Python**, ML, and backend skills flexibly across client domains, balancing data science requirements with production engineering standards and maintainable delivery practices.

Webiz Digital — Software Developer *(Jan 2014 – Sep 2016)*

- Developed **Python 2.7–3.5** and Django-based web applications for digital marketing, e-commerce, and analytics clients, focusing on backend reliability and data-driven features.
- Built early recommendation and segmentation features using pandas, NumPy, and scikit-learn 0.18–0.20 to help campaign teams personalize content and improve conversion insights.
- Created ETL scripts that cleaned CRM, website, and transaction data before loading it into PostgreSQL and MySQL reporting tables for business dashboards.

- Resolved recurring API timeout issues by optimizing database queries, adding Redis caching, and splitting slow synchronous jobs into background processing tasks.
- Migrated selected services from **Python 2.7 to Python 3.5**, fixed Unicode handling problems, updated package versions, and documented compatibility risks for future upgrades.
- Improved campaign analytics accuracy by **26%** by correcting duplicate event ingestion, adding validation checks, and reconciling backend data with dashboard metrics.
- Implemented REST endpoints for reporting, user behavior tracking, and admin workflows using Django REST Framework, PostgreSQL, Celery, and Nginx-based deployments.
- Built automated scripts for CSV processing, data imports, scheduled reports, and anomaly checks, reducing manual operations work by **30%** for internal teams.
- Supported frontend teams by delivering stable APIs, clear payload contracts, and backend documentation that made analytics features easier to integrate and maintain.
- Applied flexible Python problem-solving across backend development, data automation, and lightweight ML use cases while keeping solutions practical for client budgets and timelines.

Aurio — Junior Software Developer *(Oct 2012 – Dec 2013)*

- Supported Python and Django application development using **Python 2.7–3.3**, Django 1.4–1.6, MySQL, JavaScript, HTML, CSS, and Linux-based deployment workflows.
- Built internal admin modules, reporting pages, and backend utilities that helped operations teams manage customer records, content updates, and recurring data tasks.
- Fixed production bugs related to form validation, database constraints, broken templates, and inconsistent user input by tracing logs and improving backend error handling.
- Created Python scripts for file parsing, data cleanup, scheduled exports, and basic automation, reducing repetitive manual work for support and operations staff.
- Improved legacy SQL queries and backend views to reduce slow page loads, giving internal users faster access to operational reports and customer data.
- Helped migrate smaller modules from older Python patterns toward cleaner Django conventions, making the codebase easier for senior engineers to review and extend.
- Added unit tests for core utility functions, form workflows, and database helpers, improving confidence before releases and reducing repeated regression issues.
- Worked closely with senior developers to understand API design, deployment troubleshooting, Linux logs, and practical debugging across web and data-heavy features.
- Built a strong foundation in **Python backend engineering**, data handling, and production support that later expanded into machine learning and AI platform development.

Microsoft — Software Developer Intern *(Jun 2012 – Sep 2012)*

- Supported internal software tooling with Python scripting, SQL queries, and backend debugging while learning enterprise engineering practices, source control, and structured code review.
- Assisted senior engineers with data validation scripts, API testing, and automation tasks that improved repeatable checks across internal development workflows.
- Gained practical experience in production-quality engineering, documentation, test discipline, and communication across distributed teams working on large-scale software systems.
- Resolved early-stage bugs related to inconsistent form validation and SQL query results by reproducing issues locally, checking logs, and working with mentors to apply safer input-handling rules.

Education

Texas State University
Bachelor’s Degree in Computer Science (*Sep 2008 – May 2012)*